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HSC 4833: Lab 9 - Logistic and Count-Variable Regression

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Logistic Regression

1. **What is the model's accuracy?**
   68.42%
2. **What is the model's lowest precision?**
   0.67
3. **Do all categories have at least 30 support? (confusion matrix)**
   No. Category 0 has a support of 25 and category 1 has a support of 13
4. **Based on the pseudo-R^2 value, is this model good in this scenario?**
   The model has a McFadden's pseudo R^2 of 0.0616, which indicates a weak model fit in this scenario.

Count-Variable regression

1. **Your model should include 4 significant figures. What are they?**
   - Depression_crudeprev: p-value of 0.000
   - Selfcare_crudeprev: p-value of 0.007
   - Indeplive_crudeprev: p-value of 0.000
   - Loneliness_crudeprev: p-value of 0.000
2. **Which of these predictors is the most significant (hint: look at the coef column)**
   Indeplive_crudeprev with a coefficient of 0.1032
3. **Do any of your significant predictors have an overinflated VIF? (VIF > 10)**
   Yes. Selfcare_crudeprev with a VIF of 30.51441 and indeplive_crudeprev with a VIF of 30.953309
4. **How much of the variability is captured by the model (hint: look at pseudo R^2)**
   The pseudo R^2 is 0.04356